



AI-HED

Artificial Intelligence in Higher Education
Teaching and Learning

AI-HED.EU

BEST PRACTICE · PILOT COURSE

Digital Marketing

AI was integrated as a supporting tool for developing digital marketing plans and critically exploring case studies — combining structured team activities with reflective comparison of AI-generated and student-generated answers.



INFORMATION

MAGDALENA KUŠTEL

INTERVIEW

4 MARCH 20

INSTITUTION

FOI · ZAGREB

COURSE AT A GLANCE

A 5th-semester elective open to students across all four study modules at FOI Zagreb

4

ECTS CREDITS

61

STUDENTS

5th

SEMESTER

BA

BACHELOR LEVEL

COURSE OVERVIEW

Digital Marketing

UNIVERSITY

**University of Zagreb, Faculty of
Organization and Informatics**

PROGRAMME

Information and Business Systems

COURSE LEVEL

Bachelor · 5th semester · 4 ECTS

TYPICAL STUDENT BACKGROUND

Third-year undergraduates from all four study modules — Development of Software Systems, Artificial Intelligence in Business, Networked Systems and Computer Games, and Analysis and Design of Business Systems. The course is open to students with strong organisational and communication skills, a critical and analytical mindset, high digital literacy, and an interest in integrating information technologies into marketing strategy.

PREREQUISITES

No defined prerequisites.

SUMMARY OF AI USAGE



Artificial intelligence was primarily used as a supporting tool in developing a digital marketing plan for a new or innovative product or service. Artificial intelligence was also incorporated into seminar classes, enabling students, divided into teams max of 5 members, to explore and respond to discussion questions associated with each case study. The main objective of employing AI was to enhance students' understanding of its appropriate and effective use in digital marketing, while also promoting critical evaluation of the scope, accuracy, and specificity of AI-generated answers based on the provided prompt.

— MAGDALENA KUŠTELEGA, COURSE TEACHER

SECTION 02

02

Learning outcomes

What the course set out to teach — and where AI made a difference.

INTENDED LEARNING OUTCOMES

Three outcomes spanning strategy, technology integration, and the digital marketing mix

- 01** Propose marketing strategies for the observed company with the goal of achieving a sustainable competitive advantage.
- 02** Integrate information technologies in the marketing strategies of a company.
- 03** Identify the elements of the digital marketing mix in the context of developing a digital marketing plan for a new or innovative product or service.

AI SUPPORT · ILO 1

Covering all phases of the digital marketing plan with AI support

01**INTENDED
LEARNING OUTCOME**

Propose marketing strategies for the observed company with the goal of achieving a sustainable competitive advantage.

“

students applied AI tools to cover all phases of a digital marketing plan, with a particular focus on conducting a critical and thorough analysis of the current state. Based on this analysis, AI tools proposed appropriate marketing strategies. AI tools also assisted students in synthesizing their findings to develop a comprehensive digital marketing plan.

AI SUPPORT · ILO 2

Integrating ICT into marketing strategies through AI-assisted proposals

02**INTENDED
LEARNING OUTCOME**

Integrate information technologies in the marketing strategies of a company.

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Students utilized AI tools to integrate ICT into marketing strategies, as reflected in the digital marketing plan developed for a new or innovative product or service. They also leveraged AI to propose suitable components of the 4Ps. In seminar classes, students used AI tools to explain and evaluate key characteristics of the proposed marketing strategy for a specific company.

AI SUPPORT · ILO 3

Building the promotional mix with AI-generated creative solutions

03

**INTENDED
LEARNING OUTCOME**

Identify the elements of the digital marketing mix in the context of developing a digital marketing plan for a new or innovative product or service.

“

Students employed AI tools to identify the elements of the marketing mix when developing a digital marketing plan, with special emphasis on the promotional mix. AI was additionally used to generate creative solutions and appropriate marketing tactics, focusing primarily on the promotional aspect of the marketing mix.

SKILLS ENHANCED

Four competencies sharpened through critical AI use

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- **Critical thinking:** students had to consider and specify their search queries in order to ensure the retrieval of high-quality and relevant information.
- **Creativity:** students engaged AI to help generate new ideas and encouraged them to search for answers beyond their comfort zone.
- **Analytical thinking:** based on the queries they formulated, students learned that asking specific and simplified questions often opens up many additional questions, leading them to deepen the research process and seek answers to further issues.
- **Time management:** students became aware that the use of artificial intelligence tools does not always result in time savings, and that such tools need to be used carefully and combined with their own conclusions and reflections.

AI-SPECIFIC LEARNING OUTCOMES

AI as a support tool — not a learning outcome in itself

NO FORMAL ILO

“

Artificial intelligence is used as a supportive tool to achieve individual learning outcomes, but it is not defined as a separate learning outcome.

Artificial intelligence serves as a supportive tool for clarifying and developing specific phases of the digital marketing plan (with particular emphasis on the following plan components: analysis of the current state, operational forecasts, setting of marketing goals, marketing strategy, and marketing tactics). In seminar classes, it is used as a supportive tool for generating responses to discussion questions related to a specific business case. Additionally, the answers were given by the students without the help of AI, so that a comparison of the answers was made.

SECTION 03

03

Purpose & benefits

Why bring AI into this course — and what students and lecturer got out of it.

MAIN MOTIVATION FOR USING AI

Building awareness of AI's benefits — and responsibilities

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The primary goal of implementing AI in the course was to raise students' awareness of the benefits, as well as the responsibilities associated with using artificial intelligence tools. It also aimed to encourage student engagement and creativity while developing their critical thinking skills.

AI was used as a support tool, not as a primary data source.

BENEFITS FOR STUDENTS

From creative visualisation to critical case study analysis

“

AI tools assisted students in analysing and generating responses to discussion questions defined by a specific case study. AI was primarily also used as a supporting tool for creating visual solutions when developing a digital marketing plan for:

- Visualization of the persona profile, customer journey map, SWOT analysis, BCG matrix, landing page (based on the provided description) and so on.
- Visualization of various visuals related to individual elements of the promotional mix (e.g., posts and video content on various social networks such as Facebook, Instagram, TikTok, WhatsApp, an example of an e-mail message, flyers and so on).

Graphical representation for the 4Ps – creation of logo, brand visual, and slogan; physical display of the product; display of fixed/dynamic pricing – promotional codes, action prices.

BENEFITS FOR THE LECTURER

A new perspective on content creation and monitoring

“

AI primarily helped the lecturer by generating new, relevant ideas for preparing individual case studies linked to previously covered course material. However, it was not used as a primary data source but as a supportive tool to identify relevant global and local real-world examples. This approach boosted creativity, critical thinking, and more efficient use of time.

Since students had to attach transcripts of their AI conversations, it simplified the instructor's monitoring process to prevent AI misuse. The instructor also consulted AI to verify the originality and relevance of student-generated content.

SECTION 04

04

Implementation details

How AI showed up in class — the tools, the access model, and the guidance around them.

HOW AI WAS USED IN CLASS

Structured team activities with compare-and-reflect built in

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AI was used during seminar classes, where student teams of up to 5 members collectively employed a specific AI tool to generate responses to discussion questions tied to individual case studies. The AI was utilized across all seminar sessions for every course unit. Case study topics were diverse and closely linked to material covered in lectures that week or the previous one. But students first independently read, analysed, and responded to the questions before using AI. Students were required to complete a "Template for Using Artificial Intelligence Tools", comparing AI-generated answers to their team's conclusions on the case study. Ultimately, they addressed whether student and AI opinions differed, the extent of those differences, and whether the AI responses were suitable in the case context.

Additionally, students in teams used AI throughout the semester to develop a digital marketing plan for a new or innovative product or service, covering eight units, all addressed in lectures.

TOOLS & PLATFORMS

Seven tools spanning generative AI, research, and creative production

01

GENERAL-PURPOSE GENERATIVE AI

ChatGPT + ChatGPT 5.0

Used during seminar sessions for responding to case study discussion questions and throughout the semester for developing the digital marketing plan.

02

RESEARCH & CONVERSATIONAL AI

Perplexity Pro · MS Copilot · DeepSeek · Google Gemini · Claude AI

Used at students' discretion during seminars and plan development. Perplexity Pro was provided free of charge to FOI students in 2025/2026.

03

CREATIVE AI

Canva AI

Used for creating visual elements of the digital marketing plan — personas, promotional materials, brand visuals, and graphical representations of the 4Ps.

All tools offer free versions. Perplexity Pro was institutionally provided to FOI students. Other tools were accessed via free accounts or personal subscriptions.

ACCESS TO TOOLS - FREE VS PAID

Student choice — with institutional access to Perplexity Pro

Free

**RECOMMENDED FOR ALL
STUDENTS**

Students chose freely — free versions were the norm, with institutional Perplexity Pro available to all.

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Students had the freedom to choose any AI tool they preferred, typically opting for free ones or those they were personally subscribed to at the time.

In the 2025/2026 academic year, FOI provided free Perplexity Pro licenses to lecturers and students, so some teams exclusively used Perplexity during seminars.

GUIDANCE & SUPPORT FOR USING AI EFFECTIVELY

A structured template that embedded reflection into every session

“

Each student team was required to submit the completed "Template for Using Artificial Intelligence Tools," which consisted of four questions:

1. Which AI tool did you use in finding answers to the discussion questions?
2. Please paste here the entire conversation process with the used AI tool.
3. How many minutes in total did the interactive process with the AI tool last?
4. Compare the answers to the questions generated by the AI tool in relation to what you as a team concluded about the observed business case. Do your opinions and the given answers differ, and to what extent does that difference reach?

AI IN STUDENT ASSESSMENT

Graded for effective use — 2 points per ILO group

YES

Embedded in formal assessment criteria

“

Students earned grades not only for effectively using AI to improve their digital marketing plans but also for interpreting AI-generated content, based on these criteria:

- 2 points for AI usage out of 20 points for ILO 1 and 2 in total
- 2 points for AI usage out of 20 points for ILO 3 in total

When evaluating digital marketing plans, lecturers specifically noted the points awarded solely for AI tool utilization.

SECTION 05

05

Evaluation & risk management

How impact was measured — and how misuse was kept in check.

HOW AI IMPACT WAS EVALUATED

Weekly structured feedback and a post-course satisfaction survey

“

Student teams provided weekly feedback on the quality of AI tool responses to discussion questions (question 4 in the "Template for Using Artificial Intelligence Tools"), comparing them to their own answers and offering critical reflections explaining reasons for opinion differences. The lecturer placed special emphasis on oral commenting on this specific question, allowing various teams to freely share their views on solutions and gain insights into diverse student perceptions and final solution perspectives. The primary goal was to raise awareness of AI's possibilities and potential risks, thereby encouraging responsible use of the tool when developing digital marketing plans.

METRICS & KPIS

Post-course survey with targeted satisfaction metrics

YES

“

Upon completion of the course delivery (lectures and seminars), students were requested to complete a post-survey aimed at assessing their satisfaction and the quality of the instruction, which incorporated the implementation of an AI tool. The results of the post-survey for the Digital Marketing course are presented here.

Statement	Mean	N	Std. Dev.
This course helped me use AI responsibly, following ethical considerations and privacy rules.	3,92	38	0,912
This course helped me evaluate AI-generated answers critically.	3,81	37	1,023
This course helped me use AI effectively in the course topic(s).	3,92	38	1,194
The integration of AI in this course improved my learning experience.	3,74	38	1,201
The use of AI in this course helped me better understand complex concepts.	3,74	38	1,155
The use of AI in this course helped me apply knowledge in practical ways.	3,86	37	1,004
The teacher guided us in using AI responsibly, following ethical considerations and privacy rules.	3,97	37	1,118
The teacher guided us in evaluating AI-generated answers.	4,03	38	0,915
The teacher guided us in using AI effectively.	3,95	37	0,941
The integration with AI in this course helped me develop relevant skills for my future.	3,62	37	1,163
I feel more prepared to use AI in other academic or professional contexts because of this course.	3,66	38	1,236

FORMAL COURSE EVALUATION

Not part of the formal institutional evaluation process

NO

Outside the institution's standard process

AI use was not included in the institution's standard course-evaluation process.

RISK MANAGEMENT IN AI USE

Transparency and sequencing as the primary safeguards

YES

Rules set before activities began

“ The greatest potential risk associated with using artificial intelligence tools was linked to the fact that excessive, non-transparent, and unethical use of AI could lead to poorer learning outcomes. Initially, AI was intended to serve as a supportive tool for generating ideas and guiding student teams toward creating original solutions and responses to the posed queries and numerous challenges. In other words, the AI tool was not meant to replace student effort but rather to save time on searching numerous primary data sources, centralizing them in one place through AI.

To promote transparent, ethical AI use that enhances learning, the lecturer set these rules:

- AI permitted only after teams independently read, analysed, and answered discussion questions (seminars).
- Student teams must attach full AI conversation logs and clearly mark plan sections written/edited with AI, specifying the tool's name (seminars, digital marketing plan).

SECTION 06

06

Overall impact

What changed — for the teaching, the learning, and the student work.

OVERALL IMPACT ON TEACHING & LEARNING



The artificial intelligence tool did not have a significant impact on the teaching process, as it essentially remained unchanged.

However, using the AI tool had a significantly positive impact on the lecturer's own learning process, enabling a faster, simpler, and more creative way to generate content that should be relevant and engaging for students by its very nature. Ultimately, the lecturer gained a new perspective on formulating queries and assignments for students, with AI serving as a supportive tool that guided him toward creating specific, clear, and unambiguous prompts.

— MAGDALENA KUŠTELEGA, COURSE TEACHER

IMPACT ON STUDENT LEARNING OUTCOMES

Broader perspective and deeper critical engagement with course material

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The AI tool positively influenced the learning outcomes by helping students develop a broad understanding of the scope of AI applications, specifically in the context of digital marketing. The use of AI offered students a new perspective for examining the case study and searching for answers defined by the theme of digital marketing plan, with the primary objective of the activity focused on fostering and expressing critical thinking skills and creativity. This approach encouraged students to engage more deeply with the material, enhancing their analytical abilities and allowing them to critically evaluate both AI-generated responses and their own team's conclusions.

MAIN CHALLENGES & HOW THEY WERE MANAGED

No significant challenges — success built on clear early communication

“

Neither the lecturer nor the students encountered any potential problems or difficulties during the course of the activities, such as technical difficulties, issues with understanding AI outputs, integrating AI into work, or providing adequate support.

The main reason for the successful implementation of AI in the course lies in the fact that the lecturer timely informed students about the specific manner and sequence of activities during seminar instruction, as well as the potential consequences associated with irresponsible, unethical, and non-transparent use of artificial intelligence tools.

SECTION 07

07

Scalability & transferability

How this experience travels to other courses — and what it would take to support that.

SCALING TO OTHER COURSES AND NEED FOR INSTITUTIONAL SUPPORT

Yes — for any course with project-based or seminar assessment

100%

YES — SCALABLE

The concept can be incorporated into any course that includes project assignments or seminar instruction — particularly where AI can support idea generation, creative problem-solving, or the graphical representation of textual content.

“

The concept of work and activity implementation involving AI integration into teaching processes can be incorporated into other courses that, for example, include the obligation to complete a project assignment as part of their assessment elements. AI could be engaged in writing certain sections/chapters of the paper. For instance, AI can be used as a supportive tool for generating new ideas, stimulating creative problem-solving for posed challenges, or graphically representing textual parts of the paper. Likewise, AI can be implemented in seminar instruction to enhance students' levels of creativity and critical thinking.

KEY ADVICE FOR EDUCATORS

Three principles for purposeful AI integration

Define your purpose first.**01**

Carefully consider the primary motivation and purpose of AI implementation, and how it aligns with your defined learning outcomes before introducing it to students.

Don't over-integrate.**02**

It is not necessary to integrate AI into every assessment element — avoid overwhelming students and diluting the implementation's purpose.

Apply AI selectively.**03**

Use AI where it can best enhance the learning and teaching process. Sometimes less is more — there is no need to force AI throughout the entire semester.

A semester-long pilot that turned AI into a thinking partner — without replacing the student's own analysis.

COURSE

Digital Marketing

INFORMATION

Magdalena
Kuštelega

INSTITUTION

FOI · Zagreb

PROGRAM

Information and
Business Systems

ABOUT AI-HED

An Erasmus+ cooperation project on AI in higher education

The project "**Artificial Intelligence in Higher Education Teaching and Learning**" (AI-HED), launched with co-funding from the European Commission under the Erasmus+ Programme in the fields of education, training, youth, and sport for the 2021–2027 period, was submitted and awarded under **Key Action 2: Cooperation among organisations and institutions**.

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